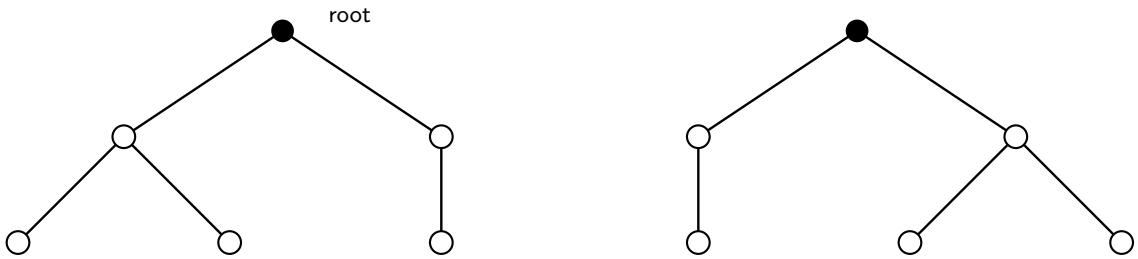
Week 12 / Catalan bijections / Grades 4–5

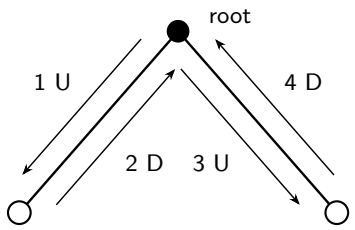
Draw dots joined by lines called edges. The filled dot at the top is the root. Every other dot hangs by one edge from exactly one dot above it. Left-to-right order matters; edge lengths and angles do not.

Two different trees with five edges:

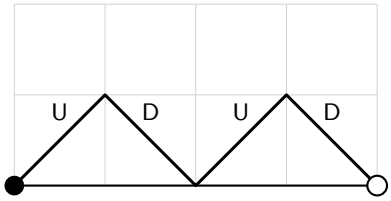


Start and finish at the root. Visit each child's whole branch from left to right before visiting the next child. Write U whenever you go away from the root along an edge and D whenever you return along it.

Example:



On the path, U goes up and D goes down.



Walk code: `UDUD`

Problem 3: Draw a tree for each code. Can a code describe two different trees?

UUDDUD

UDUUDD

UUDUDUDD

UUUDDUDD

Problem 4: Which of these codes could come from a tree? Give rules that decide for any string of U's and D's, and explain why they work.

UUUUDDDD

UUDDDUUD

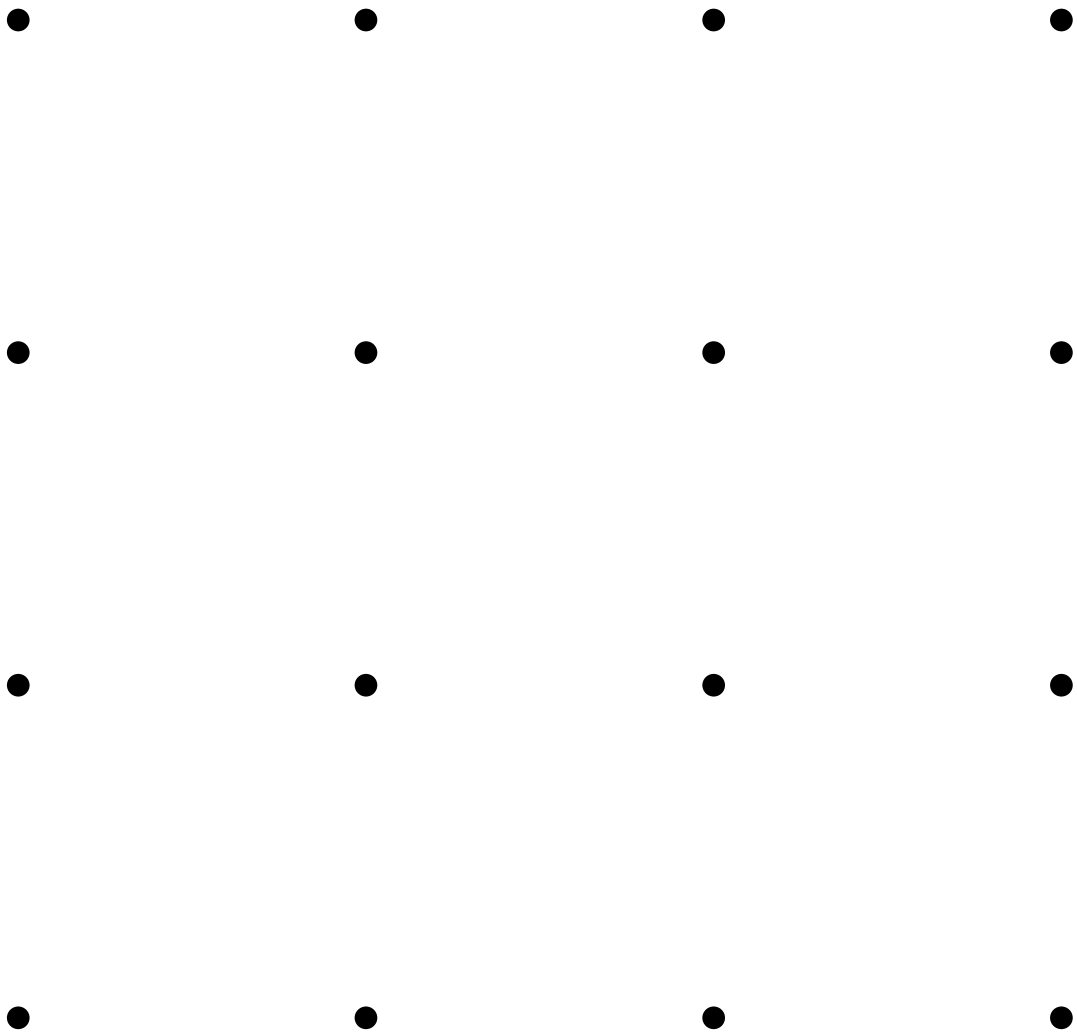
UDUDUUDD

UUDUDUDU

UUUDUDDD

DUUUDDUD

Problem 5: How many different trees have four edges? Organize the count so someone can check that every tree is counted exactly once.



Problem 6: Find the number of trees with five edges, then six edges. Give a counting rule that works for any number of edges, and explain why it counts each tree once.